

The cheapest correct answer wins — **not the smartest-sounding one.**

Track 1 is scored on Fireworks token count above an accuracy threshold — not on feature count. At hackathon scale, the difference between calling the strongest model every time and calling the right thing every time is the entire leaderboard.

The Insight

- Most tasks in a real workload don't need a frontier model.
- A meaningful fraction don't need a model at all.
- `847 × 36` is arithmetic, not a reasoning problem — sentiment classification of one sentence doesn't need chain-of-thought.
- **The win isn't a better model. It's not calling one when you don't have to — and calling the cheapest adequate one when you do.**

Architecture: **The Cascade**

STEP 0

TaskClassifier

0 tokens · regex / heuristics
decide task type before any
network call.

→

Tier 0 · deterministic solvers

0 tokens. Arithmetic (safe AST eval), date math,
string ops, unit conversion, regex extraction. Fires
only when it can self-validate.

Tier 1 · cheapest adequate model

Minimal prompt, temperature=0, reasoning
suppressed → ConfidenceGate (format validation
+ Tier-0 cross-check + logprobs secondary).
Pass → answer. Fail → Tier 2.

Tier 2 · single escalation

Escalates once to the strongest allowed model.
Never loops. No LLM self-verification.

Single escalation, never loops — a deliberate token-safety property, not a limitation. The confidence gate's primary check is deterministic (format + Tier-0 cross-check); logprobs are secondary because live probing showed the token stream mixes reasoning-channel tokens with answer tokens.

The Token-Efficiency Toolbox

- 1 Zero-token Tier 0** — deterministic solvers answer arithmetic, dates, strings, units, and extraction without touching an LLM.
- 2 Cheapest-adequate routing** — Tier 1 always picks the lowest-cost model that can handle the task's capability tag, not the biggest available.
- 3 Reasoning suppression** — per-model profiles stop models like `gpt-oss-20b` from burning completion tokens on hidden `reasoning_content`.
- 4 Minimal prompts + answer-only contracts** — ≤ 10 -token system text, per-task-type `max_tokens` caps and stop sequences.
- 5 Single-escalation confidence gate** — format + Tier-0 cross-check (logprobs secondary) before falling through to Tier 2 exactly once — never a self-verification loop.

Results — 200-task live eval

METRIC	VALUE
Accuracy	99.5% (199/200)
Zero-token tasks (Tier 0)	65/200 — 32.5%
Token reduction vs. honest baseline	~47–51%
Total price-weighted cost, 200 tasks	\$0.003636
Retry rate	2.1%

ROUTE DISTRIBUTION · 200 TASKS



Honest Limitations

- **NOT MET** **SC2 stretch target ($\geq 60\%$ token reduction)**. Measured 47–51%. The remaining cost floor is structural: gpt-oss's ~82-token fixed chat-template overhead plus a ~96–128-completion-token floor required to let its hidden reasoning channel finish without truncating.
- **One residual eval failure** (`summarization-003`): a correct paraphrase drops the literal keyword the `contains_all` grader requires. Diminishing returns to chase further on a 5-task category.
- **Gemma is not live on this Fireworks account today** — probed July 7, all Gemma variants return `NOT_FOUND` . See next slide.
- **COUNTED, NOT HIDDEN** **Retry accounting is real**: 1.45% retry rate, and retry tokens are counted in the reported totals, not subtracted out.

Gemma & the AMD Stack

- The model registry auto-prefers `gemma*` models at equal price tier — this is unit-tested and active in the code today, independent of whether Gemma is reachable.
- Live probe (July 7, real API key, `GET /models` + direct calls): all Gemma variants returned `NOT_FOUND` on this Fireworks serverless account.
- The moment a `gemma*` model is added to `ALLOWED_MODELS` — via a hackathon proxy or AMD Developer Cloud self-hosting — the registry preference activates with zero code changes.
- **We are not claiming Gemma ran in this eval. We are claiming the routing logic already knows how to prefer it.**

Team, Repo, Links

REPOSITORY

github.com/SebAustin/amd-routing-agent

Public · CI green (verified)

LIVE DEMO — HUGGING FACE SPACE

sebaustin-amd-routing-agent-demo.hf.space

Verified live 2026-07-07

Docs: `README.md`, `PLAN.md`, `SECURITY.md`, `DEPLOYMENT.md` in the repo.

Built for the AMD Developer Cloud Hackathon, ACT II — Track 1.